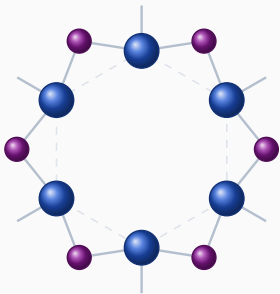


(a) Structural Graph ($\mathcal{G}$)



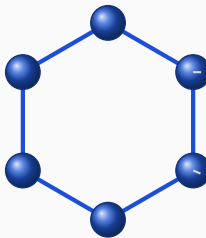
Nodes ($i \in \mathcal{V}$): All N atoms in unit cell

Edges ($(i, j) \in \mathcal{E}$): within $r_c^{\text{str}} = 8.0 \text{ \AA}$

Node Features: One-hot encoded element embeddings

Edge Features: Gaussian RBF expansion of interatomic distance

(b) Magnetic Subgraph ($\mathcal{G}_m$)



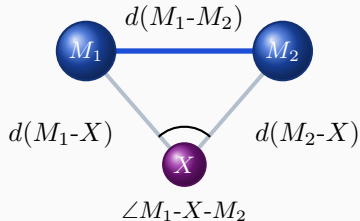
Nodes ($i \in \mathcal{V}_m$): Magnetic metal atoms only (M)

Edges ($(i, j) \in \mathcal{E}_m$): within $r_c^{\text{mag}} = 5.0 \text{ \AA}$

Node Features: Magnetic features with one-hot encoded element embeddings

Edge Features: Distance and angle-aware bond features

(c) Magnetic Edge Feature



Representations:

- Spatial coupling distances
- Fourier cosine angle expansions
- Bridging ligand features